

“I thought the free response problem sets were annoying if I’m honest. But looking back in the grand scheme of things they were so, so, helpful in identifying what I needed to review and work on.” A 2019 Nerd

FR1. 2019 Practice AP Exam (No Calculator)

Suppose f and g are continuous functions.

- a) Express, then evaluate, the following limit as a definite integral:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\left(1 + \frac{2k}{n}\right)}{\left(1 + \frac{2k}{n}\right)^2 + 4} \cdot \frac{2}{n}$$

- b) Suppose $\int_0^4 f(x) dx = 5$ and $\int_0^4 g(x) dx = -7$. Find $\int_0^4 [2f(x) - 3g(x)] dx$.

- c) Suppose $\int_{10}^{14} f(x) dx = 11$ and $\int_{10}^{20} f(x) dx = 8$. Find $\int_{14}^{20} f(x) dx$.

- d) Let $H(x) = \int_{x^2}^4 f(t)g(t)dt$. Find an expression for $H'(x)$.

FR2. 1997 BC 6 (No Calculator)

Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 24$, with the initial condition $v(0) = -50$.

- a) Find the particular solution v , in terms of t , to the differential equation with the initial condition.
- b) Terminal velocity is defined as $\lim_{t \rightarrow \infty} v(t)$. Find the terminal velocity of the skydiver.
- c) It is safe to land when her velocity is -20 feet per second. At what time t does she reach this speed?

MC1. $f(x) = \begin{cases} x & \text{for } x < 2 \\ 3 & \text{for } x \geq 2 \end{cases}$

If f is the function defined above, then $\int_{-1}^4 f(x) \, dx$ is

(A) $\frac{9}{2}$

(B) $\frac{15}{2}$

(C) $\frac{17}{2}$

(D) undefined

MC2. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = 1 + 2y$ with the initial condition $f(0) = 1$. What is the approximation for $f(1)$ if Euler's method is used, starting at $x = 0$ with a step size of 0.5?

(A) 2.5

(B) 3.5

(C) 4.0

(D) 5.5

MC3. CA Let h be the function defined by $h(x) = \frac{1}{\sqrt{x^5 + 1}}$. If g is an antiderivative of h and $g(2) = 3$, what is the value of $g(4)$?

(A) -0.020

(B) 0.152

(C) 3.031

(D) 3.152

MC4. $\int e^x \cos(e^x + 1) \, dx =$

(A) $\sin(e^x + 1) + C$

(B) $e^x \sin(e^x + 1) + C$

(C) $e^x \sin(e^x + x) + C$

(D) $\frac{1}{2} \cos^2(e^x + 1) + C$